Uniform shorter rewrites for record-atom pairs containing 1113

27 August 2026

For a positive odd integer x, let

$$U_+(x) := \frac{3x+1}{2^{v_2(3x+1)}}. \tag{1}$$

We use the four record atoms

$$A := 1, \qquad B := 12, \qquad C := 1113, \qquad D := 1122. \tag{2}$$

If a positive exponent word $w = (a_1, \dots, a_L)$ has mass

$$M(w) := a_1 + \cdots + a_L, \tag{3}$$

then its shifted affine equation has the form

$$2^{M(w)}Y = 3^L X + D(w), \qquad X := x + 1, \qquad Y := U_+^L(x) + 1. \tag{4}$$

Lemma 1 (cylinder transport)**.** *Suppose that the word w has length L, mass M, least positive ordinary representative x_0, and endpoint y_0. Every positive start with exact word w is*

$$x_k = x_0 + 2^{M+1}k, \qquad k \geq 0, \tag{5}$$

and its endpoint is

$$y_k = y_0 + 2\,3^L k. \tag{6}$$

Suppose that a word v of length $m \leq L$ and mass N is exact from z_0 to y_0. Then the same word is exact from

$$z_k = z_0 + 2^{N+1}3^{L-m}k \tag{7}$$

to y_k. In particular, if

$$z_0 < x_0, \qquad 2^{N+1}3^{L-m} < 2^{M+1}, \tag{8}$$

then $z_k < x_k$ for every $k \geq 0$.

Proof. The exact valuation cylinder of w is one odd residue class modulo 2^{M+1}, which proves (5). Substitution in (4) proves (6). The cylinder modulus for v is 2^{N+1}. Increasing its start by the quantity in (7) increases its endpoint by

$$\frac{3^m}{2^N} 2^{N+1} 3^{L-m} k = 2\, 3^L k.$$

Thus (7) reaches the endpoint in (6) with the same exact word. The final assertion follows by subtracting (7) from (5). □

Theorem 2 (complete two-atom C rewrite table)**.** *Let P be a two-atom word over $\{A, B, C, D\}$ which contains the atom C. Let x be any positive odd start having exact word P, and let y be its endpoint. Then there is a positive odd integer $z < x$ which reaches y by the shorter word displayed below.*

P	*raw source*	*alternative*	x_0	y_0	z_0
AC	11113	1121	223	425	167
BC	121113	1	411	587	391
CA	11131	11	207	395	175
CB	111312	112	463	661	391
CC	11131113	121	4815	7715	4571
CD	11131122	11122	4303	6895	3631
DC	11221113	121	3399	5447	3227

(9)

The corresponding shifted-start identities $Z = \lambda X + \eta$ are

P	λ	η
AC	3/4	0
BC	243/256	59/64
CA	27/32	1/2
CB	27/32	1/2
CC	243/256	9/16
CD	27/32	1/2
DC	243/256	21/32.

(10)

In every row the descent holds on the complete source cylinder, not only at its least representative.

Proof. Direct iteration of (1) verifies every source and alternative word at the three integers in the corresponding row of (9). It also gives the identities in (10).

It remains to extend each identity from the least representative to the complete cylinder. The source and alternative lift increments from Lemma 1 are

P	source increment	alternative increment
AC	256	192
BC	1024	972
CA	256	216
CB	1024	864
CC	8192	7776
CD	8192	6912
DC	8192	7776.

(11)

The alternative increment is strictly smaller in every row, and (9) gives $z_0 < x_0$. Lemma 1 proves the result. □

Corollary 3 (arbitrary common suffix)**.** *Let U be any positive exponent word. If PU is exact from x to q, where P is one of the seven source words in (9), replace P by its displayed alternative word and retain the suffix U. The resulting word is exact from a positive odd start $z < x$ to the same endpoint q.*

Proof. The source and alternative prefixes reach the same intermediate odd state by Theorem 2. Appending the common exact suffix U proves the claim. □

Scope. If a globally least unbounded seed begins its interior record-atom coding at the normalized origin, the first two atoms cannot contain C. At a later record state the theorem gives a smaller coalescing start relative to that record state, but it does not by itself place the new start below the original least seed. Propagating the contraction through an arbitrary earlier atom prefix remains the global arithmetic problem.

The accompanying exact certificate is

`agent-code/collatz-research/`
`verify_divergence_record_atom_pair_rewrites.py`.